

$A :=$ Taking 2 Red

$B :=$ Taking 2 Black

$C :=$ Taking 1 of each

$R :=$ Number of Red balls

$B :=$ Number of Black balls

$P(a, b) := P(R = a, B = b)$

$$P(A) = \begin{cases} \frac{\binom{R}{2}}{\binom{R+B}{2}} & \text{if } R \geq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$P(B) = \begin{cases} \frac{\binom{B}{2}}{\binom{R+B}{2}} & \text{if } B \geq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$P(C) = \begin{cases} \frac{\binom{R}{1} \cdot \binom{B}{1}}{\binom{R+B}{2}} & \text{if } R \geq 1 \wedge B \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$P(a,b) = \begin{cases} 1 & \text{if } a == 0 \\ 0 & \text{if } b == 0 \\ f(a,b) & \text{otherwise} \end{cases}$$

$$\begin{aligned} f(a,b) = & P(A; a,b) \cdot P(a-2,b) + \\ & + P(B; a,b) \cdot P(a,b) \\ & + P(C; a,b) \cdot P(a,b-1) \end{aligned}$$

$$\binom{n}{1} = n$$

$$\begin{aligned}
 P(2,2) &= P(A;2,2) \cdot P(0,2) + \\
 &\quad + P(B;2,2) \cdot P(2,2) \\
 &\quad + P(C;2,2) \cdot P(2,1)
 \end{aligned}$$

$$P(0,2) = 1$$

$$\begin{aligned}
 P(2,1) &= P(A;2,1) \cdot P(0,1) + \\
 &\quad + P(B;2,1) \cdot P(2,1) \\
 &\quad + P(C;2,1) \cdot P(2,0)
 \end{aligned}$$

$$P(0,1) = 1$$

$$P(2,0) = 0$$

$$P(B;2,1) = 0$$

$$\begin{aligned}
 \therefore P(2,1) &= P(A;2,1) \cdot 1 + \\
 &\quad + 0 \cdot P(2,1) \\
 &\quad + P(C;2,1) \cdot 0 \\
 P(2,1) &= P(A;2,1) \\
 &= \frac{\binom{2}{2}}{\binom{3}{2}} \\
 &= \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 P(B;2,2) &= \frac{\binom{2}{2}}{\binom{4}{2}} \\
 &= \frac{1}{6}
 \end{aligned}$$

$$\begin{aligned}
 P(A;2,2) &= \frac{\binom{2}{2}}{\binom{4}{2}} \\
 &= \frac{1}{6}
 \end{aligned}$$

$$\begin{aligned}
 P(C;2,2) &= \frac{2 \cdot 2}{\binom{4}{2}} \\
 &= \frac{4}{6}
 \end{aligned}$$

$$\begin{aligned}
 \therefore P(2,2) &= \frac{1}{6} \cdot 1 + \\
 &\quad + \frac{1}{6} \cdot P(2,2) \\
 &\quad + \frac{4}{6} \cdot \frac{1}{3} \\
 \equiv P(2,2) &= \frac{7}{15} = 0.4(6)
 \end{aligned}$$

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$$\begin{aligned} f(a,b) &= P(A;a,b) \cdot f(a-2,b) + \\ &\quad + P(B;a,b) \cdot f(a,b) \\ &\quad + P(C;a,b) \cdot f(a,b-1) \end{aligned}$$

$$\alpha = P(A;a,b)$$

$$\beta = P(B;a,b)$$

$$\gamma = P(C;a,b)$$

$$y = f(a-2,b)$$

$$x = f(a,b)$$

$$z = f(a,b-1)$$

$$x = \alpha \cdot y + \beta \cdot x + \gamma \cdot z$$

$$\gamma z = (1 - \beta)x - \alpha y$$

$$x = \frac{\gamma z + \alpha y}{1 - \beta}$$

$$f(a,b) = P_C \cdot f(a,b-1)$$